

# SW Engineering CSC 648/848



## Video Game Clip Competitions

### Team 01

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### Milestone 4

**November 29 2021**

#### Revision table

| Revision ID | Revision Date | Revised by                  |
|-------------|---------------|-----------------------------|
| 1           | 11/29/2021    | Damon, Kelvin, Steven, Alex |
|             |               |                             |
|             |               |                             |
|             |               |                             |

## Product Summary

For the past decade, multiplayer online video games have experienced unprecedented growth, with the global market size valued at 167.50 billion dollars in 2020. This number is expected to balloon to over 200 billion dollars at a 7.2% CAGR. This growth in video games provided a means of value creation for highly-skilled or unique players. However, this value capture had been limited to ad-revenue, sponsorships, and tournament proceeds through live streaming or live performance. We introduce our application **Pottr**, a novel monetization platform, to the internet community of content creators and gaming enthusiasts for value creation, community engagement, and offline low-risk competition. For the MVP, Pottr will feature account creation, video uploading, and voting. Future implementation is as follows:

As a user of Pottr, you can participate in community-created global tournaments for a chance to win monetary rewards. Pottr is made up of many time-limited tournaments where users can submit video game clips to any number of these ongoing competitions along with an entrance fee, which contributes to a “Competitor” prize pool. These funds are locked up and deposited into a liquidity pool to generate APR. The “Voter” pool will also be created using a similar model. Voters on Pottr are eligible to vote for submission(s), which will require some monetary backing and is contributed to a “Voter” prize pool. At the end of the voting period, the submission(s) with the largest number of votes will receive the generated APR from the “Competitor” pool, while the voters that have voted for winning submission(s) receive some proportion of the APR from the “Voter” pool. Any initial funds from users are returned, as the prize money is generated from the funds deposited into lending protocols.

While there are many esports fans, we also understand that gamers are diverse. Pottr also enables competitions that are not necessarily esports driven, but may also include other video game clips, such as fails or game breaking bugs. Pottr is a novel and exciting value creation and community engagement platform for all participants in the growing video game ecosystem.

### **Major Committed Functions:**

1. Registration
2. Login
3. Enter competitions (upload video)
4. Vote
5. Find the winner of a competition and update account balance

**URL:** <https://api.pottr.app/>

# QA testing

## Unit Test

Our focus will be on Joe the average who wants a website to be able to share his clips with other people, Mohammed the Content Creator who wants to post his amazing clips and get rewarded for them being good, and Jane the corporate who wants their game to get more attention. We will be focusing on people able to upload clips and vote for them, and being able to view certain games, and being able to know which clips got the most votes in a competition. Insomnia was used to build unit tests for testing routes and GET requests.

- **Unit Test Case categories**
  - Test to see if functions in routes/categories return the categories to the front end
  - Used Insomnia unit testing to test if the route returns status 200 and test if it returns an array of the categories
  - Passed both tests
- **Unit Test Case competitions**
  - Test if the functions in routes/competitions returns the competitions to the front end
  - Used Insomnia unit testing to see if the getAll and search functions return status 200 and array of the competition
    - Both function passed the tests
  - Test if the getByCategoryId returns an array of json containing the competitions when searching through categories
    - Functions passes the test case
- **Unit Test Case posts**
  - Test if the functions in routes/posts returns posts to the front end
  - Used Insomnia unit testing to see if getByCompId returns array of json containing the posts from a specific competition
    - Function passes the test
  - Check if getAll returns array of json containing all posts in the website
    - Function passes the test

# Integration Test

|                 |                                               |                                                                            |                                     |                          |      |                                        |  |  |
|-----------------|-----------------------------------------------|----------------------------------------------------------------------------|-------------------------------------|--------------------------|------|----------------------------------------|--|--|
| Test Case ID    | 1                                             | Test Case Description                                                      | Test the Login Cookie Functionality |                          |      |                                        |  |  |
| Created By      | Steven                                        | Reviewed By                                                                | Damon                               |                          |      |                                        |  |  |
| QA Tester's Log |                                               |                                                                            |                                     |                          |      |                                        |  |  |
| Tester's Name   | Damon                                         | Date Tested                                                                | 11/22/2021                          | Test Case(Pass/Fail/Not) | Pass |                                        |  |  |
| S #             | Prerequisite:                                 |                                                                            | S #                                 | Test Data                |      |                                        |  |  |
| 1               | Access to Chrome/Firefox Browser              |                                                                            | 1                                   | email = testy1@gmail.com |      |                                        |  |  |
| 2               |                                               |                                                                            | 2                                   | pass = 1234              |      |                                        |  |  |
| 3               |                                               |                                                                            | 3                                   |                          |      |                                        |  |  |
| 4               |                                               |                                                                            | 4                                   |                          |      |                                        |  |  |
| Test Scenario   |                                               |                                                                            |                                     |                          |      |                                        |  |  |
| Step #          | Step Details                                  | Expected Results                                                           | Actual Results                      |                          |      | Pass / Fail / Not Executed / Suspended |  |  |
| 1               | Go the the profile button icon on the website | you should get a sign up page                                              | as expected                         |                          |      | Pass                                   |  |  |
| 2               | click on sign in                              | you should be redirected to a sign in page                                 | as expected                         |                          |      | Pass                                   |  |  |
| 3               | enter the test data and click sign in         | should get a popup that says login is successful and redirect to home page | as expected                         |                          |      | Pass                                   |  |  |

|                 |                                                   |                                                                 |                                       |                          |                    |                                        |  |  |  |
|-----------------|---------------------------------------------------|-----------------------------------------------------------------|---------------------------------------|--------------------------|--------------------|----------------------------------------|--|--|--|
| Test Case ID    | 2                                                 | Test Case Description                                           | Tests to see if uploading video works |                          |                    |                                        |  |  |  |
| Created By      | Steven                                            | Reviewed By                                                     | Damon                                 |                          |                    |                                        |  |  |  |
|                 |                                                   |                                                                 |                                       |                          |                    |                                        |  |  |  |
| QA Tester's Log |                                                   |                                                                 |                                       |                          |                    |                                        |  |  |  |
|                 |                                                   |                                                                 |                                       |                          |                    |                                        |  |  |  |
| Tester's Name   | Damon                                             | Date Tested                                                     | 11/25/2021                            | Test Case(Pass/Fail/Not) |                    | Pass                                   |  |  |  |
|                 |                                                   |                                                                 |                                       |                          |                    |                                        |  |  |  |
| S #             | Prerequisite:                                     |                                                                 |                                       | S #                      | Test Data          |                                        |  |  |  |
| 1               | Access to Chrome/Firefox Browser                  |                                                                 |                                       | 1                        | a 87 mb video file |                                        |  |  |  |
| 2               | video game clip to upload                         |                                                                 |                                       | 2                        |                    |                                        |  |  |  |
| 3               |                                                   |                                                                 |                                       | 3                        |                    |                                        |  |  |  |
| 4               |                                                   |                                                                 |                                       | 4                        |                    |                                        |  |  |  |
|                 |                                                   |                                                                 |                                       |                          |                    |                                        |  |  |  |
| Test Scenario   |                                                   |                                                                 |                                       |                          |                    |                                        |  |  |  |
|                 |                                                   |                                                                 |                                       |                          |                    |                                        |  |  |  |
| Step #          | Step Details                                      | Expected Results                                                | Actual Results                        |                          |                    | Pass / Fail / Not Executed / Suspended |  |  |  |
| 1               | login using valid account                         |                                                                 |                                       |                          |                    |                                        |  |  |  |
| 2               | go to the website and click on the compete button | should go to an upload page                                     | as expected                           |                          |                    | Pass                                   |  |  |  |
| 3               | choose what game to enter                         | should display what game you chose                              | as expected                           |                          |                    | Pass                                   |  |  |  |
| 4               | choose what competition to enter                  | should say which competition you chose inside the game category | as expected                           |                          |                    | Pass                                   |  |  |  |

|   |                                                                              |                                                                                 |             |      |
|---|------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------|------|
| 5 | enter the title of video and choose to enter pot                             | should just show what title you put in and whether you want to enter the pot    | as expected | Pass |
| 6 | click on upload video and choose a video to upload from your computer system | should show the clip name you want to upload                                    | as expected | Pass |
| 7 | click on upload                                                              | should take a few seconds and upload the clip under the competition you entered | as expected | Pass |
| 8 | go back to competition to check if posts it there                            | see your recently uploaded post there                                           | as expected | Pass |

## Code Review

- **Coding Style**

- For coding style we mostly adhered to Google's coding style (such as camelcase for variables), and would enforce it as time permitted through code review. We would also use prettier to apply uniform styling to the entire codebase. This would be checked with a GitHub Action before being merged into "master".

- **Code Review**

- Code review was performed before merging from working branches into the "production" branch. Code review was done through GitHub where the person with the code would create a pull request and request one or two reviewers to look at the code. Reviewers would add comments, and if there were major errors, request changes. Otherwise, once the review is approved, the code is merged into the main branches for GitHub.

## Non functional Requirements

| <b>Non functional requirement</b>                       | <b>Status</b>   |
|---------------------------------------------------------|-----------------|
| The application will run on mobile and desktop browsers | <b>Done</b>     |
| The user's password will be encrypted                   | <b>On Track</b> |
| All data shall be stored in AWS EC2                     | <b>Done</b>     |
| The application will target a USA audience              | <b>Done</b>     |